Frequently Used Notation

$f^{-1}(A)$	the inverse image or preimage of A under f
$a \mid b$	a divides b
(a, b)	the greatest common divisor of a, b also the ideal generated by a, b
$\lvert A\rvert, \lvert x\rvert$	the order of the set A, the order of the element x
$\mathbb{Z}, \mathbb{Z}^+$	the integers, the positive integers
$\mathbb{Q}, \mathbb{Q}^+$	the rational numbers, the positive rational numbers
$\mathbb{R}, \mathbb{R}^+$	the real numbers, the positive real numbers
$\mathbb{C}, \mathbb{C}^\times$	the complex numbers, the nonzero complex numbers
$\mathbb{Z}/n\mathbb{Z}$	the integers modulo n
$(\mathbb{Z}/n\mathbb{Z})^\times$	the (multiplicative group of) invertible integers modulo n
$A \times B$	the direct or Cartesian product of A and B
$H \leq G$	H is a subgroup of G
Z_n	the cyclic group of order n
D_{2n}	the dihedral group of order $2n$
S_n, S_Ω	the symmetric group on n letters, and on the set Ω
A_n	the alternating group on n letters
Q_8	the quaternion group of order 8
V_4	the Klein 4–group
$\mathbb{F}_N$	the finite field of N elements
$GL_n(F), GL(V)$	the general linear groups
$SL_n(F)$	the special linear group
$A \cong B$	A is isomorphic to B
$C_G(A), N_G(A)$	the centralizer, and normalizer in G of A
$Z(G)$	the center of the group G
G_s	the stabilizer in the group G of s
$\langle A \rangle, \langle x \rangle$	the group generated by the set A, and by the element x
$G = \langle \ldots \mid \ldots \rangle$	generators and relations (a presentation) for G
$\ker \varphi$, $\operatorname{im} \varphi$	the kernel, and the image of the homomorphism φ
$N \trianglelefteq G$	N is a normal subgroup of G
gH, Hg	the left coset, and right coset of H with coset representative g
$\lvert G : H\rvert$	the index of the subgroup H in the group G
$\mathrm{Aut}(G)$	the automorphism group of the group G
$Syl_p(G)$	the set of Sylow p-subgroups of G
n_p	the number of Sylow p-subgroups of G
$[x, y]$	the commutator of x, y
$H \rtimes K$	the semidirect product of H and K
$\mathbb{H}$	the real Hamilton Quaternions
$R^\times$	the multiplicative group of units of the ring R
$R[x], R[x_1, \ldots, x_n]$	polynomials in x, and in $x_1, \ldots, x_n$ with coefficients in R
RG, FG	the group ring of the group G over the ring R, and over the field F
$\mathcal{O}_K$	the ring of integers in the number field K
$\varinjlim A_i, \varprojlim A_i$	the direct, and the inverse limit of the family of groups A_i
$\mathbb{Z}_p, \mathbb{Q}_p$	the p-adic integers, and the p-adic rationals
$A \oplus B$	the direct sum of A and B

$LT(f)$, $LT(I)$	the leading term of the polynomial f, the ideal of leading terms
$M_n(R)$, $M_{n\times m}(R)$	the $n \times n$, and the $n \times m$ matrices over R
$M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$	the matrix of the linear transformation φ with respect to bases $\mathcal{B}$ (domain) and $\mathcal{E}$ (range)
$\operatorname{tr}(A)$	the trace of the matrix A
$\operatorname{Hom}_R(A, B)$	the R-module homomorphisms from A to B
$\operatorname{End}(M)$	the endomorphism ring of the module M
$\operatorname{Tor}(M)$	the torsion submodule of M
$\operatorname{Ann}(M)$	the annihilator of the module M
$M \otimes_R N$	the tensor product of modules M and N over R
$\mathcal{T}^k(M)$, $\mathcal{T}(M)$	the kth tensor power, and the tensor algebra of M
$\mathcal{S}^k(M)$, $\mathcal{S}(M)$	the kth symmetric power, and the symmetric algebra of M
$\bigwedge^k(M)$, $\bigwedge(M)$	the kth exterior power, and the exterior algebra of M
$m_T(x)$, $c_T(x)$	the minimal, and characteristic polynomial of T
$\operatorname{ch}(F)$	the characteristic of the field F
K/F	the field K is an extension of the field F
$[K : F]$	the degree of the field extension K/F
$F(\alpha)$, $F(\alpha, \beta)$, etc.	the field generated over F by α or α, β, etc.
$m_{\alpha,F}(x)$	the minimal polynomal of α over the field F
$\operatorname{Aut}(K)$	the group of automorphisms of a field K
$\operatorname{Aut}(K/F)$	the group of automorphisms of a field K fixing the field F
$\operatorname{Gal}(K/F)$	the Galois group of the extension K/F
$\mathbb{A}^n$	affine n-space
$k[\mathbb{A}^n]$, $k[V]$	the coordinate ring of $\mathbb{A}^n$, and of the affine algebraic set V
$\mathcal{Z}(I)$, $\mathcal{Z}(f)$	the locus or zero set of I, the locus of an element f
$\mathcal{I}(A)$	the ideal of functions that vanish on A
$\operatorname{rad} I$	the radical of the ideal I
$\operatorname{Ass}_R(M)$	the associated primes for the module M
$\operatorname{Supp}(M)$	the support of the module M
$D^{-1}R$	the ring of fractions (localization) of R with respect to D
R_P, R_f	the localization of R at the prime ideal P, and at the element f
$\mathcal{O}_{v,V}$, $\mathbb{T}_{v,V}$	the local ring, and the tangent space of the variety V at the point v
$\mathfrak{m}_{v,V}$	the unique maximal ideal of $\mathcal{O}_{v,V}$
$\operatorname{Spec} R$, $\operatorname{mSpec} R$	the prime spectrum, and the maximal spectrum of R
$\mathcal{O}_X$	the structure sheaf of $X = \operatorname{Spec} R$
$\mathcal{O}(U)$	the ring of sections on an open set U in $\operatorname{Spec} R$
$\mathcal{O}_P$	the stalk of the structure sheaf at P
$\operatorname{Jac} R$	the Jacobson radical of the ring R
$\operatorname{Ext}_R^n(A, B)$	the nth cohomology group derived from Hom_R
$\operatorname{Tor}_n^R(A, B)$	the nth cohomology group derived from the tensor product over R
A^G	the fixed points of G acting on the G-module A
$H^n(G, A)$	the nth cohomology group of G with coefficients in A
Res, Cor	the restriction, and corestriction maps on cohomology
$\operatorname{Stab}(1 \trianglelefteq A \trianglelefteq G)$	the stability group of the series $1 \trianglelefteq A \trianglelefteq G$
$\|\|\theta\|\|$	the norm of the character θ
$\operatorname{Ind}_H^G(\psi)$	the character of the representation ψ induced from H to G